Ashwin Conrad

Calgary, AB

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Mechanical engineering student with hands-on experience in vehicle low-voltage systems, industrial panel assembly, engineering analysis, and technical automation. Interested in automotive, motorsport, manufacturing, test, and product-development engineering roles.

EDUCATION

University of Alberta — Mechanical Engineering - Edmonton, AB — **2023 - Present**

WORK EXPERIENCE

AltaGas - Calgary, AB | Engineering Co-op — **2025 - 2026**

- Developed Python, Microsoft Fabric, and Power BI workflows for corrosion monitoring, inspection planning, and asset lifecycle work across multi-facility operations.
- Analyzed 2+ years of operating data for 12+ brazed aluminum heat exchangers, screening against OEM limits and summarizing monitoring gaps and risk-based recommendations.

Spartan Controls - Calgary, AB | Control Panel Technician — **Summer Terms, '23, '24**

- Built, wired, loaded, and checked 30+ industrial control panels using engineering drawings, electrical schematics, pre-delivery QA, and documented deficiency resolution.
- Processed 250+ PLC software updates in under half the allocated time while supporting panel production and quality checks.

LEADERSHIP	COMMUNITY INVOLVEMENT
Formula EV Racing - Low Voltage Electronics Lead Lead low-voltage electrical-system planning and integration for Formula UA27, coordinating board-level hardware, safety interfaces, sensors, power distribution, and vehicle packaging across mechanical and software teams.	**CYDC Basketball Coach** Youth coaching and team development.
Engineering Student Engagement Leader 2024-2025	

GENERAL SKILLS

Mechanical Design	**Project Management**	**Electrical Panel Design**
Low Voltage Electronics	**Data Analysis**	**Fabrication & CAD**

RECOGNITION

Undergraduate Student Leadership Award, 2024-2025	**Leadership in Engineering Scholarship, 2023-2024**	**APEGA Science Olympics Medalist: Silver, 2021-2023**

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PROJECT PORTFOLIO

SELECTED PROJECT PORTFOLIO

Heat Exchanger Lifecycle Analysis | **AltaGas** — **2025 - 2026**
Python, data analysis, technical assessment
Analyzed 2+ years of operating data for 12+ brazed aluminum heat exchangers to assess fatigue risk and monitoring gaps. Summarized lifecycle findings into practical recommendations for inspection planning and operating awareness.

Inspection Scope Generation Tool | **AltaGas** — **2025 - 2026**
Python, openpyxl, python-docx
Developed a Python tool to generate 500+ equipment-specific inspection scopes from database exports. Reduced manual document creation time by standardizing Excel-to-Word scope generation.

Formula EV Electrical Systems | **Formula EV Racing** — **Winter 2025 - Present**
Low-voltage electrical systems
Coordinate low-voltage power distribution, sensor integration, and safety-interface planning around the UA27 vehicle package. Work across mechanical and software constraints to make electrical systems accessible, serviceable, and competition-ready.

Leather Tool Carrier | **Spartan Controls** — **2024**
Shop workflow tooling
Designed and built a leather tool carrier to keep common hand tools organized and accessible during panel assembly. Turned a personal workflow problem into a durable shop aid for faster repeated work.

TECHNICAL SKILLS

CAD & Design	SolidWorks, AutoCAD & 2D laser cutting, engineering drawings, mechanical prototyping, 3D modelling & printing
Vehicle & Electrical	low-voltage systems, power distribution, sensor integration, electrical schematics, panel assembly, QA
Data & Automation	Python, Excel data analysis, Power BI, Microsoft Fabric, Power Query, openpyxl, python-docx
Engineering Software	Aspen HYSYS, Maximo, JDE
Fabrication	Metal, leather, composite, and small mechanical build projects

Project images, technical drawings, and expanded case studies available online. **ashwinconrad.ca**